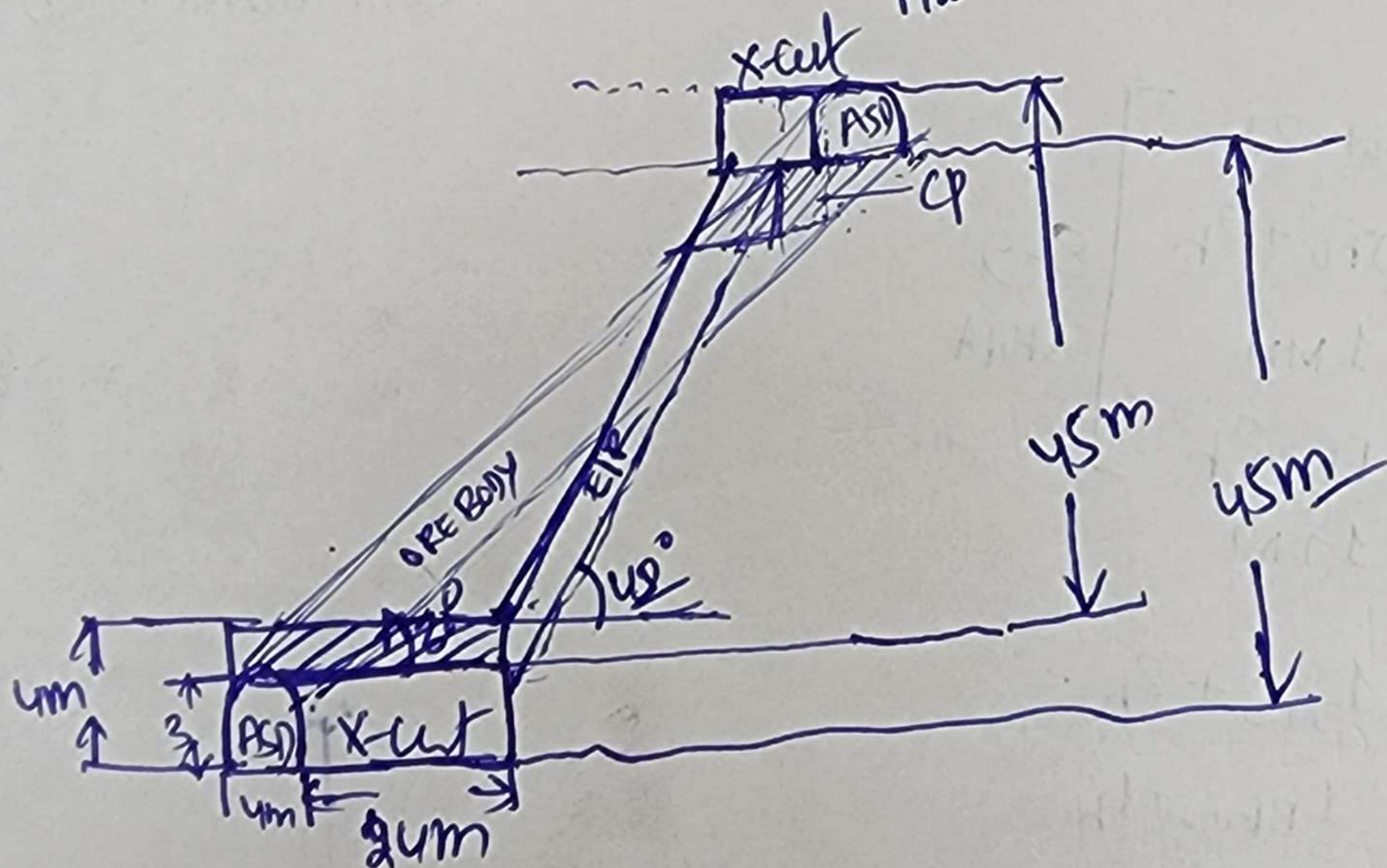
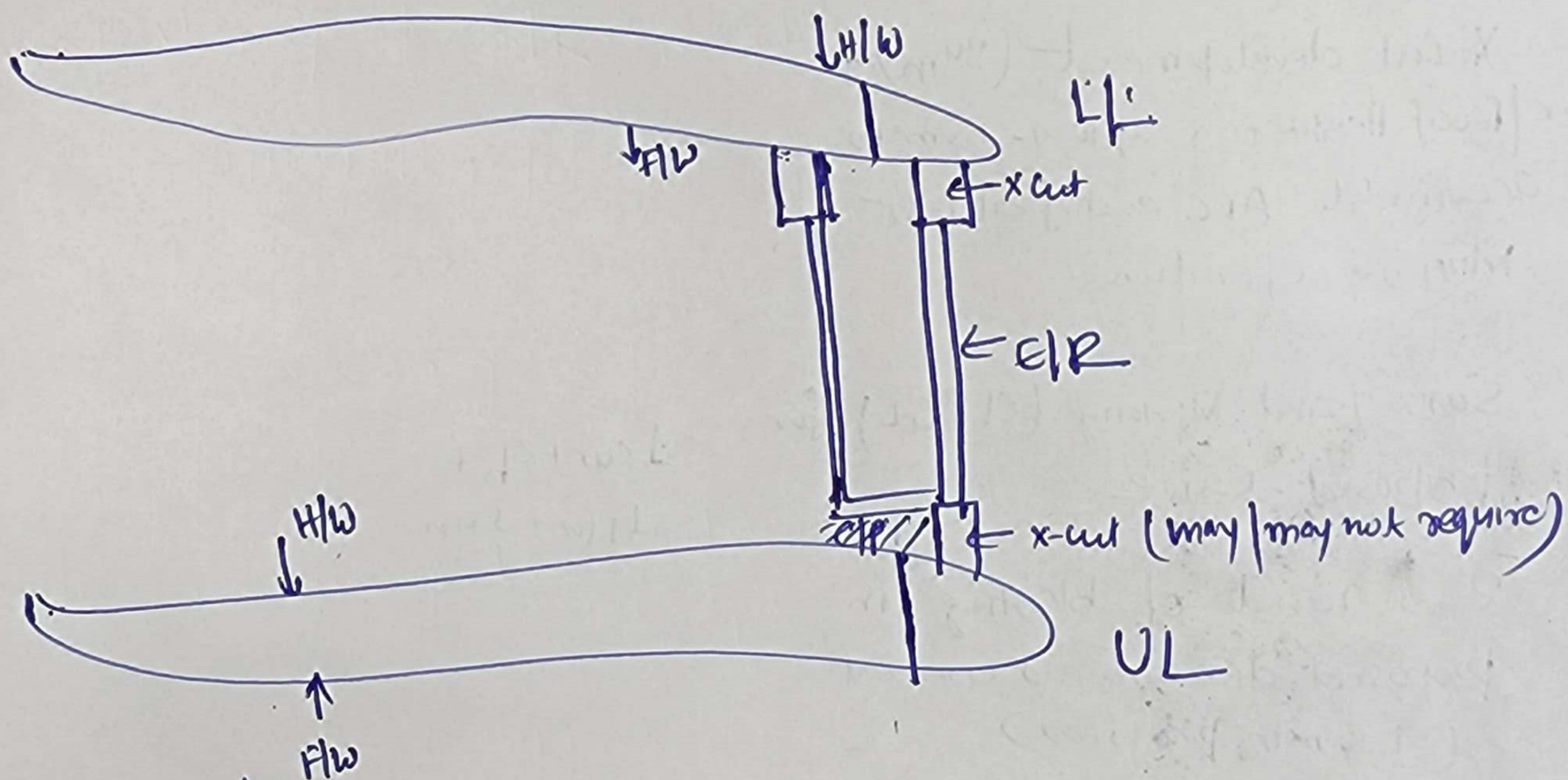
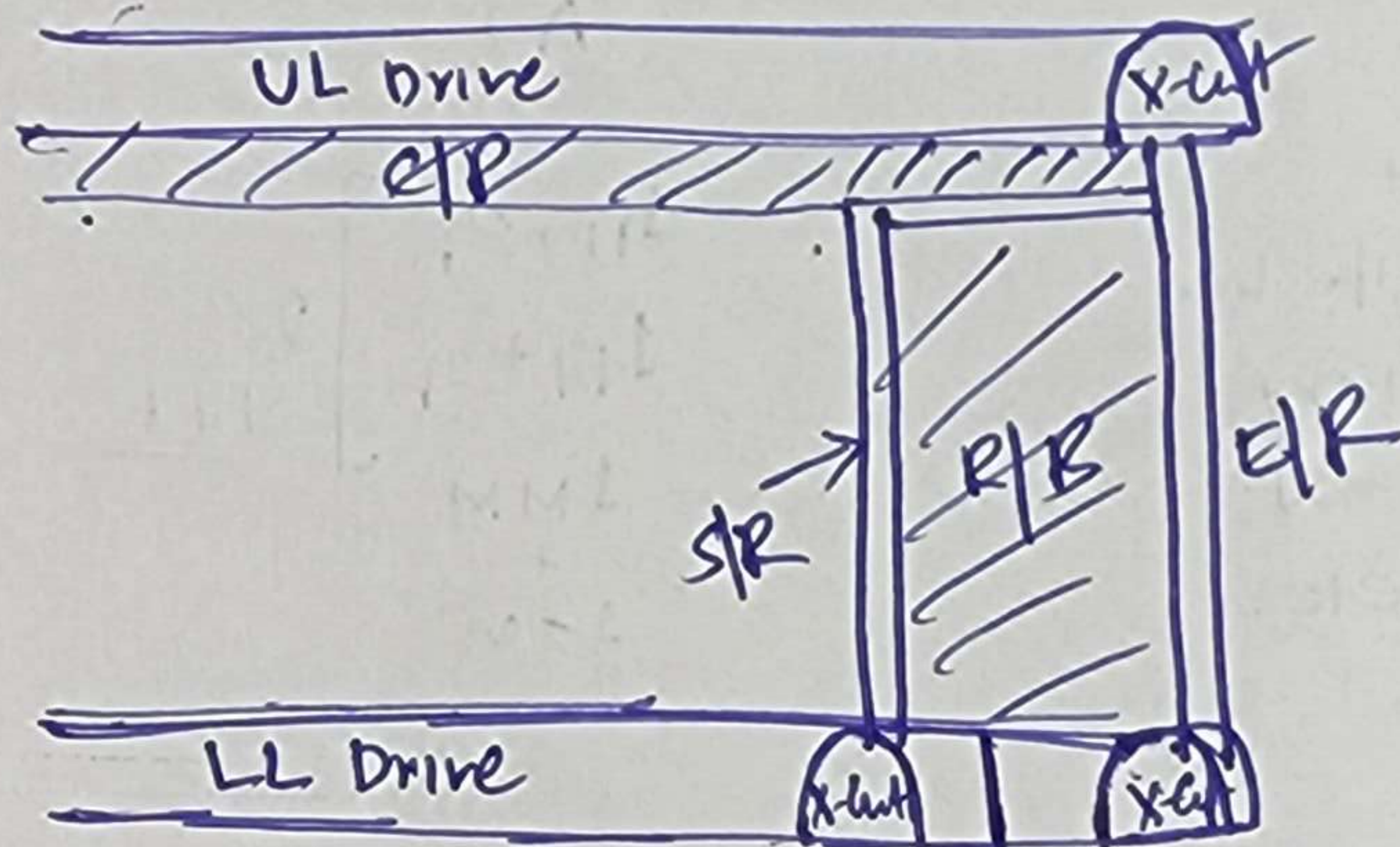
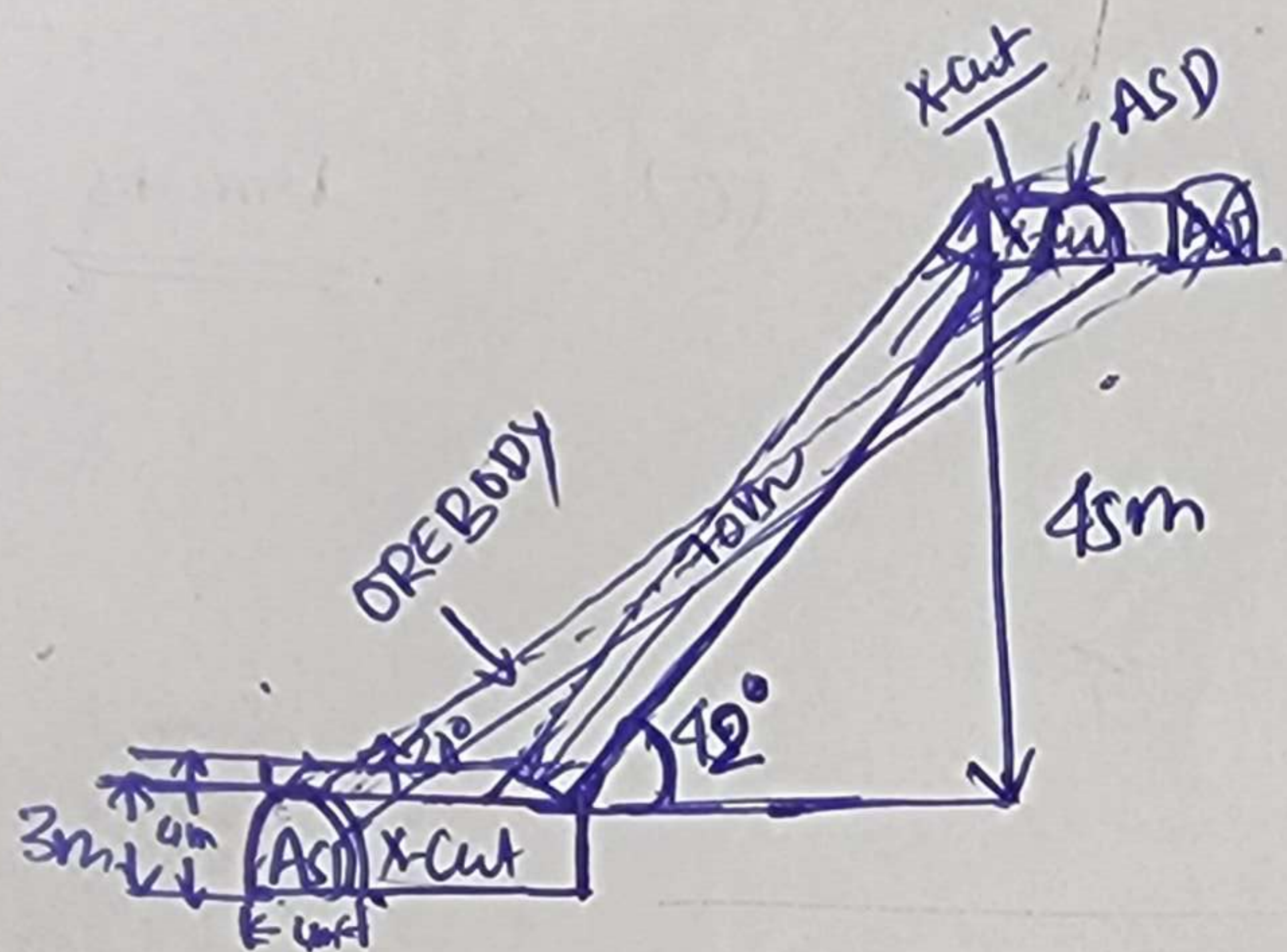


For 70m AR Raise development using ARC in given Condition :



Solution 4 (b) (i)

Steps in Installation:

- (*) Angle guide fixing, Service guide fixing, Cage fixing, Motor fixing, Hose reel Setup, Connection of hose reel with Motor. etc., Platform fixing, Grappling.
- (1) ~~These~~ preparatory work (extra of water pipe line compressed air pipeline, GEB, Duct pipe etc)

Steps

- Planning & Survey
- Survey + CL/GL Marking for X-cut (if required)
- X-cut Development (say 10m) + Roof Ht. to 4m to accommodate ARC and facilitate Mucking operation
- Installation of ARC (2)
- Survey + CL/GL for proposed Raise
- 2-3 round of blasting in proposed direction and Gradient
- Installation of ARC (3)
- Drilling face drilling + charging & setting
- Face clearance + W&L + face preparation + Repetition of cycle for 2-3 round of blasting
- Service Guide Rail extension to face & 2m length + face drilling (Incomplete)

(*) Service Guide is fixed with the help of Anchor bolt at 2 feet.

- ↳ face drilling is completed in next shift
- ↳ Repetition of cycle including Guide. Service guide Rail extra.
- ↳ Mucking after 3-4 round of blasting
- ↳ Development Survey for CL and GL atn. along with measurement of opening size after 15-20m ^{face} advancement
- ↳ Uninstallation of ARC

↳ fitting of ^{wooden} Ladder in the floor to make it operational.

Manpower distribution plan :

Activity	Shift (A)	(B)	(C)	Remarks
1. Preparatory work Water pipe line + Compressed air pipe line Vent. duct + Auxiliary Fan Installation + GEB + lighting etc.	$\left. \begin{array}{l} 1PF+2h \\ 1EI+2h \\ 1MM \\ + \\ 1TM \end{array} \right\} \begin{array}{l} 2-3 \\ \text{Shift} \end{array}$			
2. Survey and Marking (CL&GL) for X-cut	$1sur+2h+1TM+1MM$			
3. X-cut development (24m) + back/Roof Heightening upto 4-4.5m to accommodate ARC and facilitate Mucking operation				Follow manpower distribution plan for Drift/Drive
4. Survey and Marking (CL&GL) for proposed Raise	$1sur+2h+1TM+1MM$			
5. 2-3 rounds of blasting ^(6') in proposed direction and Gradient (4-5 mtr. progress)				Follow manpower distribution plan as discussed in Conventional Raising
6. Shifting & Installation of ARC	$\left. \begin{array}{l} 1Met+2h \\ 1Jho+2h \\ 1MM \\ 1PF+2h \\ 1TM \end{array} \right\} \begin{array}{l} 8-9 \\ \text{Shift A} \end{array}$			
7. Face drilling + Charging + Blasting + FC FC + WS&LD + FP.	$\begin{array}{l} 1Jho+2h \\ 1Blaster+1h \\ 1MM+1TM \end{array}$			
8. FC + WS&LD + FP + Face drilling (Incomplete)	$\begin{array}{l} 1TM+1MM \\ + 1Jho+2h \end{array} \rightarrow$			
9. Face drilling + Charging & Blasting (Complete)	$\begin{array}{l} 1Jho+2h \\ + 1Blaster+1h \\ + 1MM+1TM \end{array}$			
Repetition				
10. Service Guide Rail extrn. + Face drilling (Incomplete)	$\begin{array}{l} 1Jho+2h \\ 1TM+1MM \end{array}$			Repetition of Step 7, 8, 9 for 2-3 rounds of blasting.

4/6

	A	B	C	Remarks
11. Face drilling complete + Charging & Blasting + Fe.		1 JH + 2h 1 Bk + 1h 1 TM + 1 MM		
12. W&L LD + FP + Face drilling (Incomplete)		1 TM + 1 MM + 1 JH + 2h		
13. Face drilling (Complete) + Charging & Blasting + Fe		1 JH + 2h 1 Bk + 1h 1 TM + 1 MM		
14. Same W&L LD + FP + Service Guide extn + Face drilling (Incomplete)		1 TM + 1 MM 1 MM + 1 JH + 2h		Repetition of step 10, 11, 12 & 13
15. Mucking after 3-4 round of blasting		—		
16. Development Survey (CL & GL extn + Opening Size Measurement) after 15-20m Face progress / development		1 Sur + 2h		
17. Uninstalling of ARE		1 Mech + 2h + 1 TM + 1 MM	15 shift	
18. Fitting of Wooden ladder		1 TM + 2h + 1 MM + 1h	15 shift	

⇒ Total days required to complete it

Activity	Shift	Activity	Shift
1	2-3 shift	7, 8, 9, 10	4 blast in 1st shift
2	1 shift	11, 12, 13	7 shift
3	12 shift		(Considering 2 shift working in Raising before)
4	1 shift		(50 days for 65m development)
5	3 of 3 shift	15.	HR
6	8-9 shift	16.	1 shift
		17.	15 shift
		18.	15 shift

81

Total days required = 81 days for 70 m development

So, 50, days required for 1m development = $\frac{81}{70} \approx 1.28 \text{ days/m}$

Stope Block Development and its Completion

Q 5) (a) Types of the excavation:

- 1) Initial X cut which will act as drift later on.
- 2) cross X cut
- 3) Development of end raise/ventilation raise
- 4) Development of drift (ASD, advanced strike drift) (100 m)
- 5) Drift development (100 m)
- 6) ASD development (next 100 m)
- 7) next end raise/ventilation raise
- 8) drift development (next 100 m)
- 9) E/R - Drift - X cut
- 10) 2 Stope raises have to be developed (5 m below level)
- 11) ramp up development (after 2-3 rounds of slicing)
- 12) development of ore transfer raise.

9

(ii) Days required:

Type of excavation
in sequence

Days
required

1) initial X cut (40 m,
4.5 m X 3.2 m)

$$\frac{40}{2.9} + \frac{10}{100} \times \frac{40}{2.9}$$

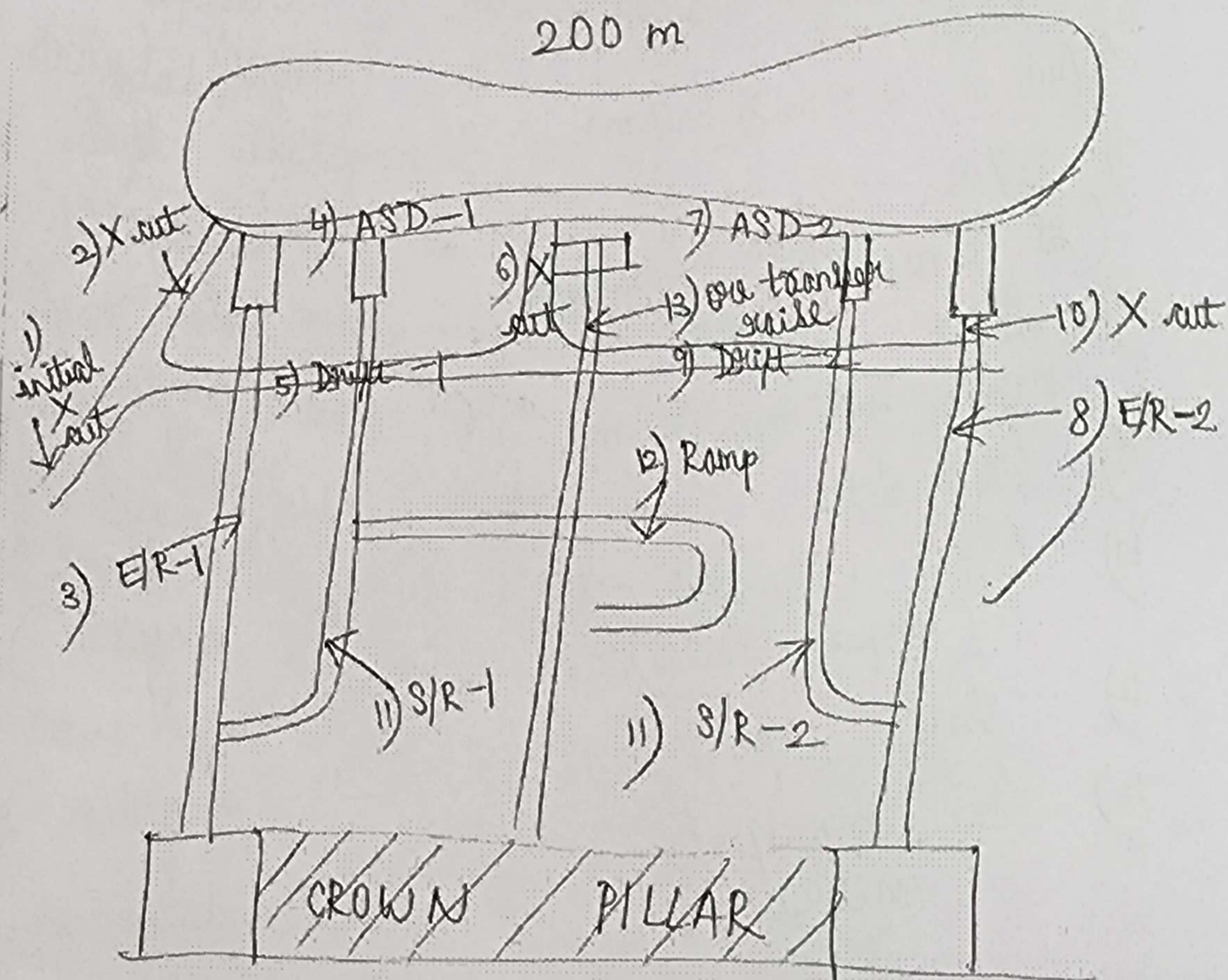
20 days

- 2) End nail - 1
(70 m, 45°, 2m X 2m) 100 days
- 3) ASD - 1
(100 m, 4.5m X 3.2m) 40 days
- 4) Drift - 1
(100 m, 4.5m X 3.2m) 40 days
- 5) well X cut
(35 m, 4.5m X 3.2m) 20 days
- 6) ASD - 2
(100 m, 4.5m X 3.2m) 40 days
- 7) E/R - 2
(70 m, 45°, 2m X 2m) 100 days
- 8) Drift - 2
(100 m, 4.5m X 3.2m) 40 days
- 9) X cut 20 days
- 10) S/R - 1 & S/R - 2
(slope nail) 90 days
- 11) Ramp up 20 days
- 12) Ore transfer
nail 100 days

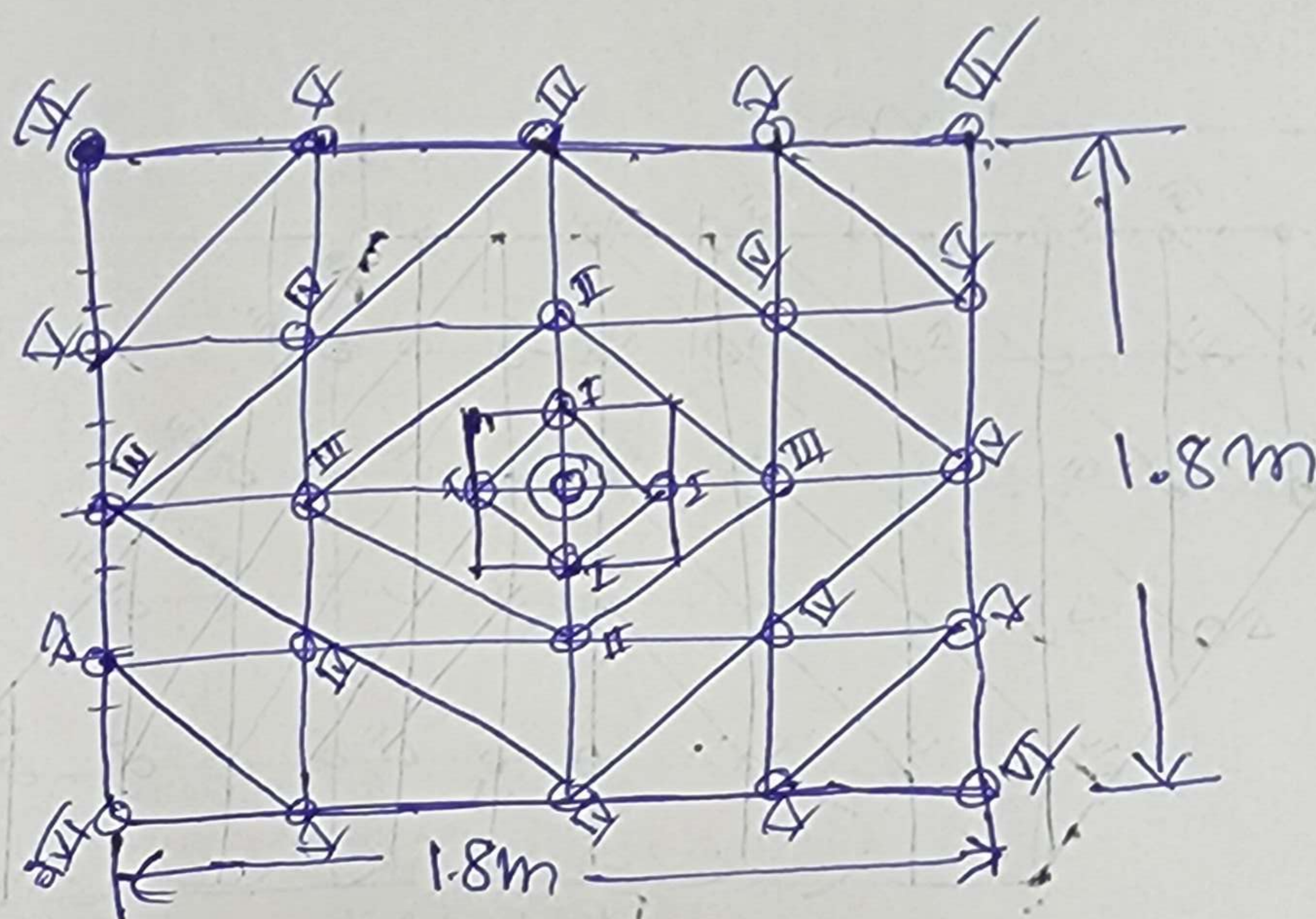
Total time = 2 ~~months~~ years

∴ we need to consider double the time
because there may be chances of

unforeseen circumstances and breakdown of
machineries,
hence total time required for completing
block development will be 4 years.



Drilling & Blasting Pattern of Jack Hammer Face (2m x 2m) : Raising (Conventional & ARC)



Note: Roman letter shows sequence of blasting

$$\text{Total No. of Holes} = 29 + 1R$$

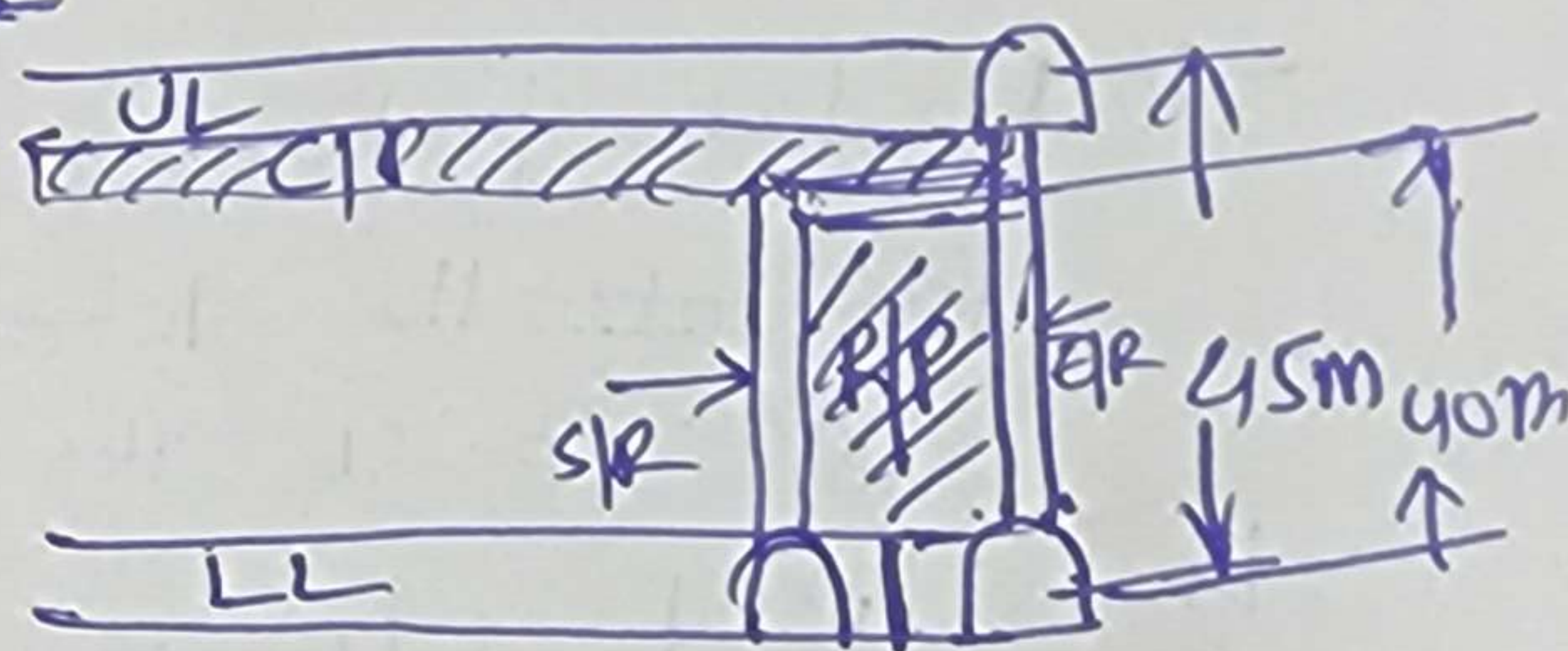
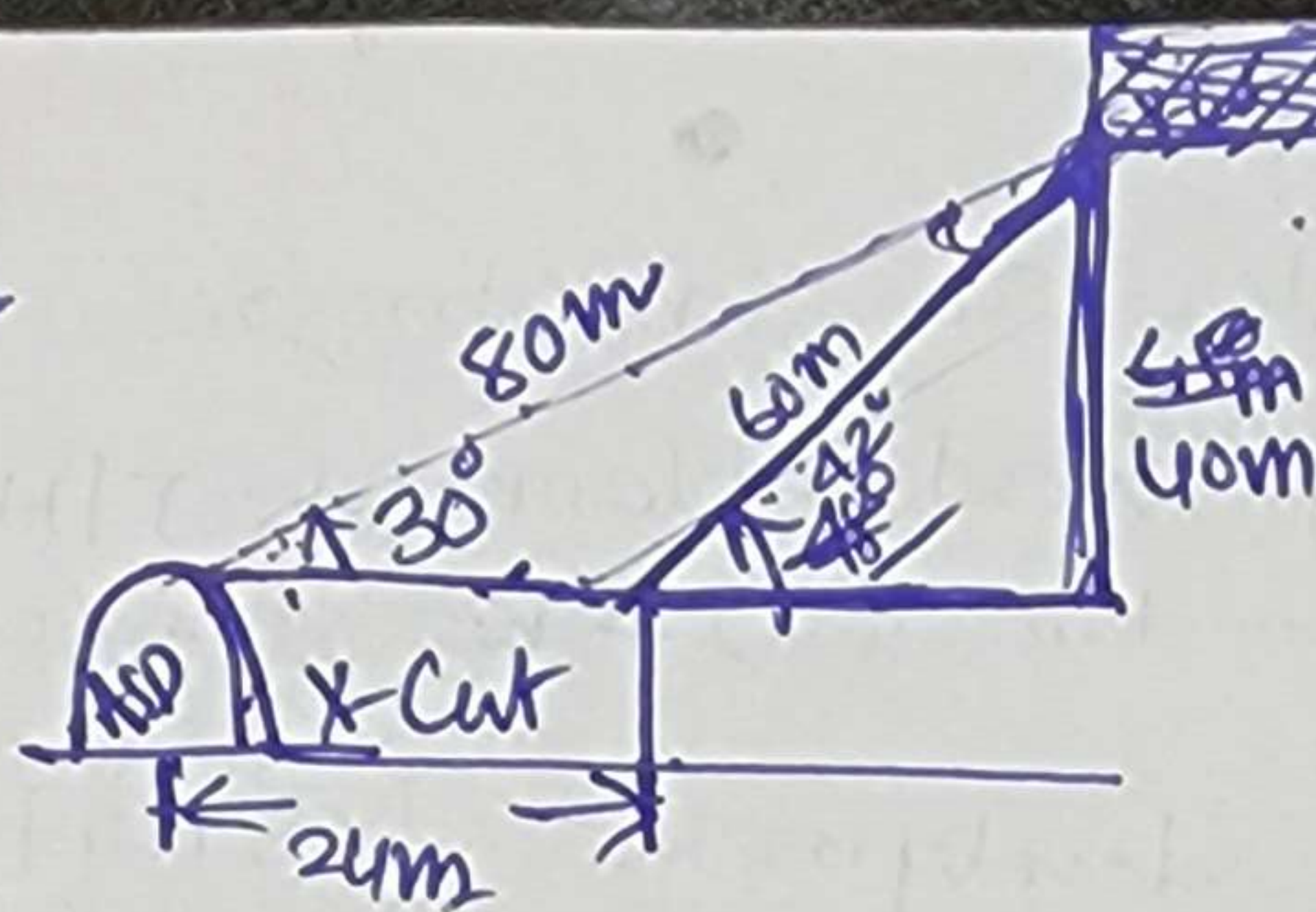
$$\text{Total No. of Blast holes} = 28$$

$$\text{Total Motorage (4 feet per hole)} = \frac{28 \times 1.2}{\text{depth of}} = 29 \times 1.2 + 1 \times 1.2$$

$$\text{Total Explosive per round} = 28 \times 7 \text{ kg} = 20 \text{ kg}$$

$$\text{Total Tonnage per round of Blasting} = 2 \times 2 \times 1.2 \times 9 \times 3 = 13 \text{ tonnes.}$$

CONVENTIONAL RAISING



① Steps Involved:

1. Preparatory Work : Compressed Air pipe line, water pipe line, Auxiliary fan + vent. duct, lighting. etc
2. Survey and Marking of S/R X-cut (if required)
3. Development of S/R Xcut in proposed direction as marked by Surveyor.
4. Survey and Marking of S/R for direction and Gradient.
5. ^{Face} Drilling using J/H drill m/c with $2\frac{1}{4}$ " & 6" ~~rod~~ drill rodⁱⁿ (Burn cut pattern)
6. Charging & Blasting
7. Fume clearance
8. Water spraying + Loose dressing + face preparation
9. Repeatability of steps 5, 6, 7, 8 for next three round of blasting
10. ^{partial} Mucking using LHD/LPDT &
11. Survey + Marking + fixing pegs
12. Platform preparation for next round of drilling ^{at least} 2m behind the face. + Rope ladder fixing in the floor
13. ^{+ side support holes for platform + ladder} ~~Face~~ Drilling + Charging & Blasting (after dismantling the platform) + ~~face~~ fume clearance + WS + LD + Face preparation + Platform preparation
14. After 2-3 rounds of blasting ^{partial} Marking is done.
15. Development Survey after 10-15m progress : checking C/L & G/L and extending C/L and G/L in the proposed direction + measuring the face size (Height and width)
16. extending the Rope ladder as face advances (Generally it is ^{kept} just behind the platform position.)

- Manhole preparation after 20-25 m progress of the face ~~to~~ for ~~facilitate the~~ Keeping wooden planks + JH U/C (Face Marking + Drilling + C&B (after platform dismantling) + ~~W&L~~ FC + W&L + Face preparation)
- S/R to E/R X-cut development (5 meters) in Horizontal plane
- # Survey and Marking C/L & G/L in proposed direction and Gradient
 - # Next 2-3 round of blasting with the help of platform
 - ## Rest without platform
 - # Muck Scraping is an additional step in the X-cut development ~~which~~ which takes more time for face preparation for next round of drilling & blasting)

- (B) Manpower distribution plan Activity wise : SN
- (C) Days required to Complete the Raise : SN 92 days
- (D) Equip/Machinery deployed : SN
- (E) Drilling and Blasting pattern : From ARC Note
- (F) Drilling Mtrge per round of blasting : From ARC Note
- (G) Explosive Consumption per round of Blasting : from ARC Note
- (H) Tonnage produced per round of blasting : from ARC Note
- (I) Powder factor obtained : from ARC note
- (J) Ventilation ckt for ventilating the face : from ARC Note

Development of Horizontal excavation (Such as: Drift | Drive | X-Cut | etc) ^{using} Drilling and Blasting excavation Method :

↳ ~~Planning~~ Survey of the proposed location(s) as per the plan and ~~for~~ Marking Central line and Grade line (Direction and Gradient)

↳ Drilling with drill jumbos or Jack hammer in the proposed direction and gradient (Drilling pattern: Burn cut for hard rock, Dia of hole Length of hole 3.2m (drill jumbo); 8 feet ²m (Jack hammer), face size 4.5m x 3.2m or 3x3m

↳ Charging & Blasting : Charging with emulsion explosive in the form of Cartridge, Electric delay detonators are used, Blasting with the help of exploder. from a safe distance atleast one right angle from the face direction. Blasting is done at the end of the shift. during Blasting Auxiliary fan should remain OFF.

↳ ~~fume clearance~~ :

↳ fume clearance : fume clearance is done with the help of Auxiliary forcing fan | Combination of forcing & exhaust fan.

↳ Water Spraying & Loose Dressing : Water Spraying & Loose Dressing is done Manually. Water Spraying suppresses the dust and loose dressing operation helps in separating loose rock generated after blasting in the back and sides of the face.

↳ ^{Hauling} Loading of Blasted material : Loading ^{operation} ~~at face~~ is carried out using LHD & LPDT combination in Trackless Mining System whereas Tub | Grizzly Car & Rippers Shovel | Crawler loader (pneumatically operated) used in Track bound Mining System. Loaded Material is dumped near by u/g Void or (if available) otherwise hoisted to surface and dumped in ~~the~~ waste dump yard.

↳ face preparation | ~~Ready~~ Ready for next round of drilling : finding sockets and Misfire (if any) and cleaning the holes.

↳ Support of back & side (if required) : Support of back as per the Support plan (SSR) framed by the Mine with the approval of DGMS. Generally, Roof bolt of 2m is used to support the back in the grid of 1.2m x 1.2m. (Cement + sand + water) is used as a grouting material. full column grouting is done.

↳ Repetition of the process (Drilling → Charging & Blasting → fume clearance → water spraying & loose dressing → Loading & Hauling of blasted rock → ^{face ready} Support)

↳ Development Survey :

fixing Centre peg & Grade peg :— In order to maintain direction and Gradient of the excavation / Development type, after 15-20m face advancement around 20-25m (In the case of Drill jumbo face and around 10-15m (In the case of Jack hammer face), Centre pegs and Grade pegs are fixed by the Surveyors in the back and sides of face respectively. ^{a set of new established} Centre pegs (2 Nos) and Grade pegs (2+2=4 Nos) set is ~~shifted~~ after 20-25m face advancement.

Development Survey :

Measuring Size of the opening : (H x W) In order to assess the size of the opening as per the actual size, Development Survey is done. It helps in monitoring the overbreakage or underbreakage. (both are undesired)

↳ Drainage : Drainage is to be provided to drain the water from the face. (For Decline and Ramp down, face pump is required during drilling and charging). For drift / drive, it is developed in up-gradient (1 in 200) ^{and drain is required}. For Ramp up and Ramp down, No need of drainage / drain. For Decline: Drainage is made at the ^{one} side of the floor.

↳ Extension of Auxiliary Services : Water pipe line, Compressed air pipe line, Electrical cable, Gate end box, Manhole Ventilation duct. are extended at regular Interval as face advances.

→ Extension of Auxiliary Services : Water pipe line, Compressed air pipe line, Electrical cable, Gate end box, Manhole, Ventilation duct, ^{peg, set up, loading bay} are extended at regular interval as face advances.

<u>Other activities</u>	<u>No. of Shift required</u>
→ Survey and Marking	1 shift
→ Preparatory Work : Compressed Air pipe line, Water pipe line Installation of Auxiliary fan + ^{duct} Electrical Cable Laying, GEB.	3 shifts
→ Loading bay	11 shift
→ Manhole + Support holes	1 shift @ 20m, for 100m . 5 shift
→ Peg Set up + size Opening size measurement	1 shift @ 20m, for 100m. 5 shift

Main Activities

Drilling + charging
+ ^{use of} ~~use of~~ ^{time clearance} + blasting + Loading &
Hauling

3 shift per round of blasting

Support (back + side)

• 2 shift @ every 3 round of blasting

$$\Rightarrow \text{Total Shift needed (for 100m development)} = \frac{100}{2.88} \cdot \frac{100}{3.2 \times 0.9} \times 3 + \frac{100}{3.2 \times 0.9} \times \frac{2}{3} + 1 + 3 + 11 + 5 + 5$$

$$= 104 + 23 + 25 = 152$$

$$\Rightarrow \text{Total Shift after taking time Bld and unforeseen situations} = 152 + 152 \times 5\%$$

$$= 152 + 7.6 = 159.6 \approx 160$$

$$\text{No. of days} \rightarrow \text{In days, } \frac{160}{3} = 53 \text{ days } \mp \frac{160}{3} =$$

$$\rightarrow \text{In month, } \frac{53}{25} = 2.12 \text{ months}$$

Table 14.6 Comparison of Advantages and Disadvantages of Underground Methods

Characteristic	Unsupported			Supported			Caving		
	Room-and-Pillar	Stope-and-Pillar	Shrinkage	Sublevel	Cut and Fill	Stull	Square-Set	Longwall	Sublevel Caving
1. Mining cost	20% Large	10% Large	45% Moderate	20% Large	55% Moderate	70% Small	100% Small	15% Large	15% Large
2. Production rate	High	High	Low	High	Moderate	Low	Low	High	Moderate
3. Productivity	High	Moderate	Low	Moderate	Moderate	Low	Low	High	Moderate
4. Capital investment	Rapid	Rapid	Rapid	Moderate	Moderate	Rapid	Slow	Moderate	Moderate
5. Development rate	Limited	Limited	Limited	Moderate	Moderate	Limited	Unlimited	Moderate	Moderate
6. Depth capacity	Low	High	Moderate	Low	High	High	High	Low	Low
7. Selectivity	Moderate	Moderate	High	Moderate	High	High	Lowest	High	High
8. Recovery	Moderate	Low	Low	Moderate	Low	Low	High	Low	High
9. Dilution	Moderate	High	Moderate	Low	Moderate	High	High	Low	Moderate
10. Flexibility	Moderate	High	High	High	High	Moderate	High	High	Moderate
11. Stability of openings	Moderate	Low	Low	Low	Low	Moderate	Low	High	High
12. Subsidence	Good	Good	Good	Good	Moderate	Moderate	Poor	Good	Good
13. Health and safety	Highly mechanized, caves with pillar recovery, good ventilation	Mechanized, fair ventilation	Gravity flow in stope, intensive, simple method	Gravity flow in stope, mechanized, large blasts, good ventilation	Gravity flow in stope, mechanized, requires backfill	Gravity flow in stope, labor-intensive, simple method	Gravity flow in stope, labor-intensive, high timber cost	Highly mechanized, continuous, rigid, expensive moves	Mechanized, draw control critical
14. Other									Gravity flow in undercut, low breakage cost, good ventilation, draw control critical

Table 14.7 Selection Chart for All Underground Mining Methods

Deposit Shape	Deposit Orientation	Deposit Thickness	Ore Strength	Rock Strength	Applicable Method(s)
Tabular	Horizontal, flat	Thin	Strong	Strong	Room-and-pillar mining, stope-and-pillar mining, Longwall mining
		Thick	Weak, strong Strong Weak, strong	Weak Strong Weak	Stope-and-pillar mining, Sublevel caving
	Vertical, steep	Thin	Strong	Strong	Shrinkage stoping, sublevel stoping
			Strong	Weak	Cut-and-fill stoping, square-set stoping, stull stoping
		Thick	Weak Weak Strong	Strong Weak Strong	Square-set stoping, Square-set stoping, Shrinkage stoping, sublevel stoping
			Strong	Weak	Cut-and-fill stoping, sublevel caving, square-set stoping
Massive			Weak	Strong	Sublevel caving, block caving, square-set stoping
			Weak	Weak	Sublevel caving, block caving, square-set stoping
			Strong	Strong	Shrinkage stoping, sublevel stoping
			Weak	Weak, strong	Sublevel caving, block caving, square-set stoping

Sources: Modified after Peele, 1941; Lucas and Haycocks, 1973; Thomas, 1978.

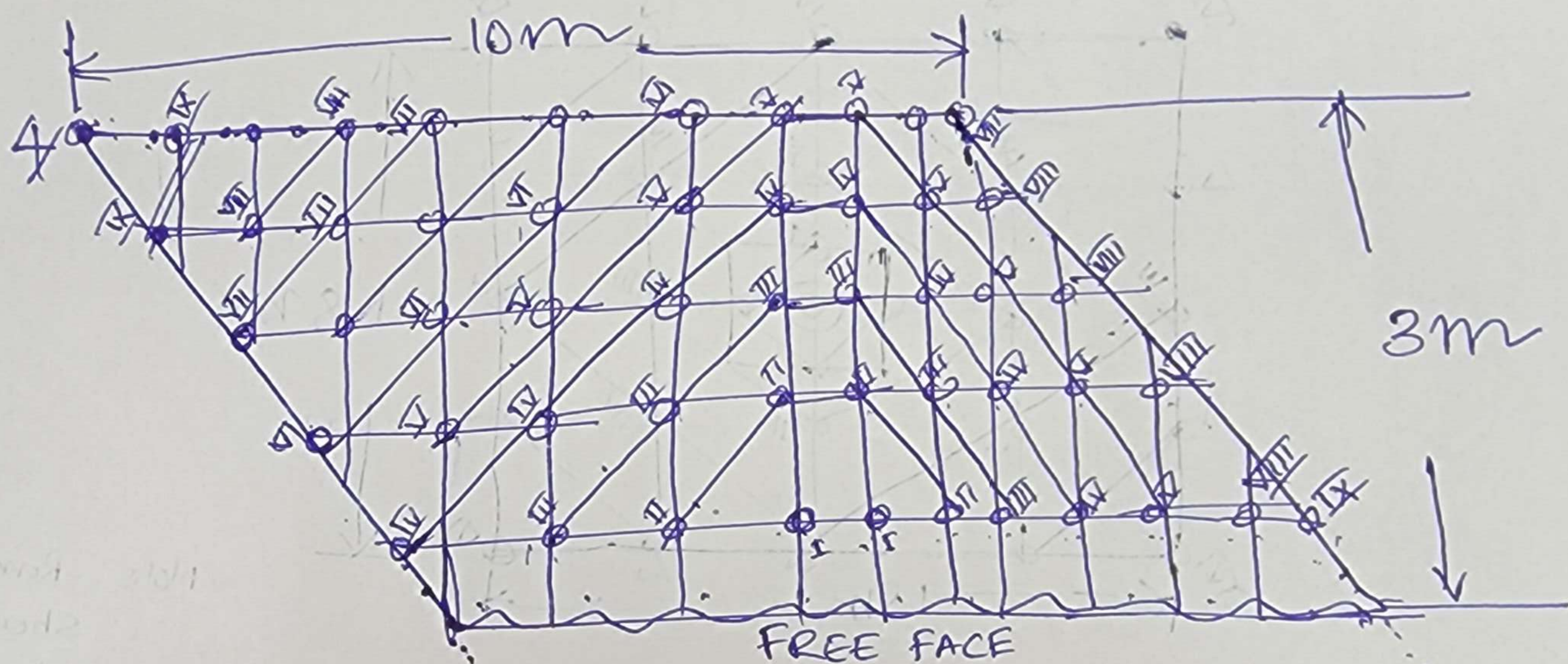
14.2.3 Novel Versus Traditional Methods

Because their characteristics and applications differ significantly, it is more difficult to compare traditional methods with novel methods than surface with underground methods. By their very nature, novel methods have been devised to meet the unique requirements of a particular mineral deposit or to employ the unusual features of a new technology. In addition, they may have been utilized less than traditional methods and may not have a suitable track record for cost estimation. Accordingly, they will ordinarily be more risky than traditional mining methods.

The characteristic of the mineral deposit that is most discriminating in the selection of a novel mining method is the mineral commodity itself (see Table 13.1). Accordingly, we summarize the most likely possibilities for each of a series of deposit types:

Slice drilling and Blasting pattern (HCF)

~~Face size~~



Total No. of Blast holes ≈ 50

Total Mrgt per round of blasting $\approx 50 \times 3.2 =$

Explosive per hole ≈ 2.7 to 3 kg

Total Explosive per round of blasting $\approx 50 \times (2.7-3) =$
 $\approx 150 \text{ kg}$

Total Tonnage per round of Blasting $= 10 \times 3 \times 3.2 \times 3$
 $\approx 270 \text{ tonnes}$

Note: Roman Letter shows sequence of Blasting.

Sl. No. 9

③ Level interval is 45m, Dip of the Orebody is 30°, thickness (Plan) 9m. etc

A block of HCF of length 200m ^{in between} two levels

particular level is to be developed using with a combination of Trackless and track based Mining Systems. Describe the following:

- 1) Types of excavation required in its development
- 2) Steps involved in its development
- 3) Time required to complete the block development
- 4) plan / sections / Drawing, wherever needed.

5)

Sl. No. 11

A developed block of 200m for HCF is to be stoped with trackless mining system; plan width of the orebody is 9m. Dip = 30°, Level interval 45m.

Describe the following:

- 1) Steps involved in the stoping process
- 2) Time required for its completion
- 3) Manpower distribution plan shift wise
- 4) plan / sections / figure wherever needed
- 5) Drilling ^{nos per round} ^{combination} of blasting
- 6) Explosive per round of blasting
- 7) Drilling pattern
- 8) Blasting pattern

examine your condition, wherever needed.

1 Steps

~~of the~~

→ planning and survey (stake plan preparation)

→ updated plan with Generation of ^{QCMS} ~~change the~~ strike

→ Slope plan drive back ^{3.2 m to 5.0 m}

→ Heightening of roof (drive from 4.2m to 3.0m across the strike)

→ Ht + widening up to the length of the block (both side ~~to the strike~~ due to strike)

→ Ht. of Slope Raise

→ Slope Survey and marking of stowing line up to 3m

→ back filling (waste rock + mill tailing)

→ Ht + widening up to the length of the block (other side up to S/R) along the strike

→ Ht. of Slope Raise

→ Slope Survey and marking of stowing line

→ Back filling (waste rock + mill tailing)

→ Roof Ht. of Access cut and marking ^{ht} ~~stake~~ ^{to access the slope}

→ Staking (slicing) of 3m with 9m width along the strike of Subblock-1

→ along the strike up to S/R

→ Back fill of Subblock-1

→ Stop Survey and marking of stowing line

Date:

Annex (RPS) 24/11/2023

Integration of 3D-Numerical Modelling
and Finite Elements for developing
a ~~finite element~~ land surface deformation prediction
Model.

~~Mining induced land surface deformation
Prediction Model~~

↳ Design of literature Review

↳ Slicing of sub-block-2 up to 1st of
3m and width 9m along strike
up to S/R

↳ Back filling of sub-block-2
↳ ~~stop~~ Survey and marking of stowing in

↳ Repeatability of process upto the green
pillars

↳ Access of slope from ramp ~~area~~
for next round of slicing (After
every three round slicing Access
would be from next Access Xcut
from the ramp up

Unit operation in per round of blasting

↳ Slicing
Drilling → charging & Blasting →
frame clearance → water spraying water →
loading & hauling → ~~blasting~~ support
drilling (back and wall) + roof bolting
(back + wall)

X X X